

Species Interactions

Introduction to Species Interactions

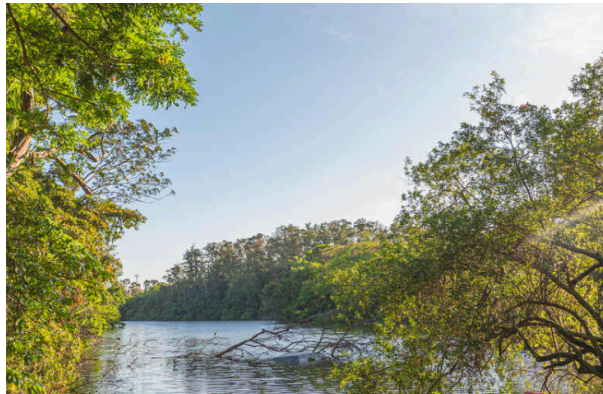
Species interactions are fundamental to ecology, influencing population dynamics, community structure, and evolution. All organisms participate in consumer-resource relationships, forming the basis of food chains and ecosystems.

Types of Species Interactions

Interactions are classified by effects on each species: benefit (+), harm (-), or neutral (0).

Main types include:

- Consumer-resource interactions (+/-): Predators benefit, prey suffer
- Competition (-/-): Both species harmed by resource sharing
- Mutualism (+/+): Both species benefit
- Commensalism (+/0): One benefits, other unaffected
- Amensalism (-/0): One harmed, other unaffected



The prickly pear cactus example demonstrates biological control through predator introduction, showing how consumer-resource dynamics can regulate invasive populations.

Consumer-Resource Interactions

All non-photosynthetic organisms consume others, creating fundamental ecological relationships that drive energy flow through ecosystems.

Categories of Consumers

- **Predators:** Capture and consume prey, removing them from populations (e.g., lions)
- **Parasites:** Feed on living host tissues without immediate killing
- **Parasitoids:** Insect larvae that consume hosts, causing eventual death
- **Herbivores:** Plant consumers, acting as predators or parasites
- **Detritivores:** Consume dead organic matter (commensal relationship)

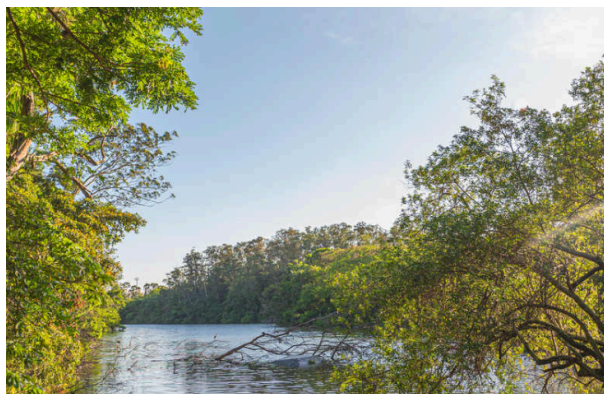


Predators like African lions specialize in capturing large prey, relying on stealth and power rather than speed.

Classification by Duration and Intimacy

Consumer-resource interactions vary by association length and resource mortality probability:

- Short/casual: Grazers, browsers (low mortality)
- Long/close: Parasites, many herbivores (variable mortality)
- Parasitoids: High mortality despite intimate association



Parasitoid wasps develop within host insects, consuming tissues and causing death after pupation.

Evolutionary Dynamics of Consumer-Resource Interactions

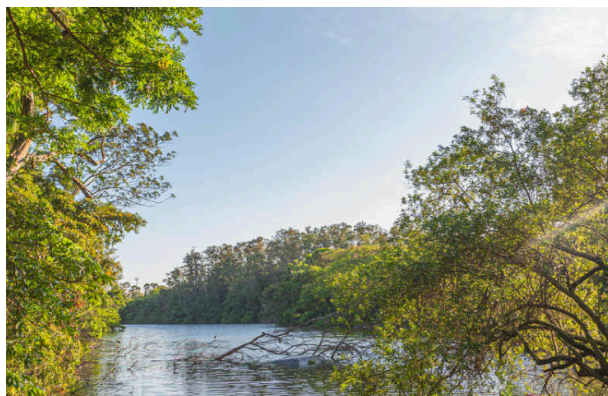
Resource and consumer populations co-evolve, with prey developing defenses and predators countering them.

Prey Defense Strategies

- Behavioral: Hiding, refuge use, fleeing
- Chemical: Toxins, repellents (e.g., bombardier beetle sprays)
- Structural: Spines, armor, tough tissues



Predator avoidance in frog tadpoles reduces activity and growth, demonstrating trade-offs between safety and development.



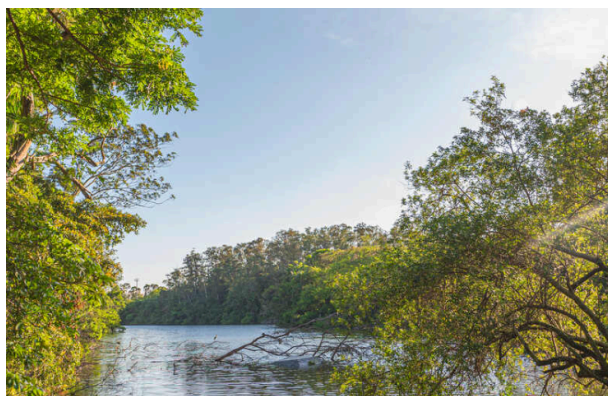
Chemical defenses like those of bombardier beetles impose costs but provide protection against predators.

Parasite-Host Relationships

Parasites maintain delicate balances with hosts, often involving complex life cycles and immune evasion.

Parasite Characteristics

- Smaller than hosts, internal or external
- Reduce host fitness without immediate killing
- Complex dispersal strategies between hosts

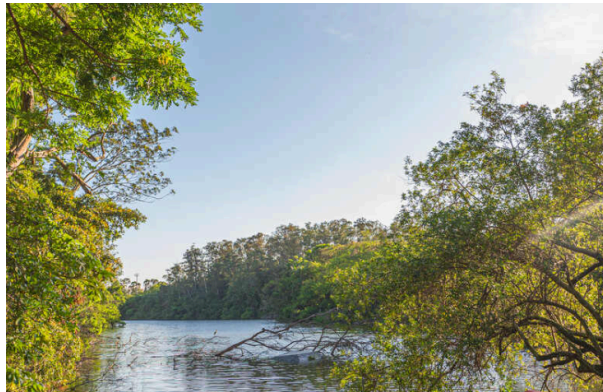


Wolbachia bacteria parasitize insect cells, manipulating reproduction and demonstrating potential evolution toward mutualism.

Parasite Life Cycles

Many parasites require multiple hosts and environmental stages, as in malaria:

- Vector transmission (mosquitoes)
- Host invasion and proliferation
- Immune evasion through surface changes



Plasmodium's life cycle involves alternation between mosquito and human hosts, with specialized stages for dispersal and infection.

Host Defenses and Parasite Virulence

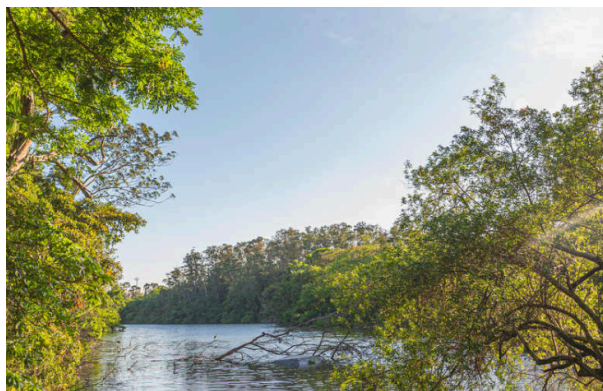
- Host immune responses: Antibodies, inflammation
- Parasite countermeasures: Surface mimicry, immune suppression
- Balance between virulence and transmission

Herbivory and Plant Defenses

Plants defend against herbivores through nutritional and chemical strategies, with costs and trade-offs.

Plant Defense Mechanisms

- Low nutritional quality: Cellulose, tannins
- Secondary compounds: Alkaloids, terpenoids, phenolics
- Structural defenses: Spines, resins, tough tissues



Cacti and milkweed use spines and toxic latex to deter herbivores, illustrating structural and chemical defenses.

Constitutive vs. Induced Defenses

- Constitutive: Always present, costly when herbivory rare
- Induced: Activated by damage, more efficient



Cotton plants induce defenses against mites, reducing pest populations and suggesting cost-benefit trade-offs.

Indirect Interactions and Competition

Species interactions create indirect effects across trophic levels and through resource sharing.

Exploitation Competition

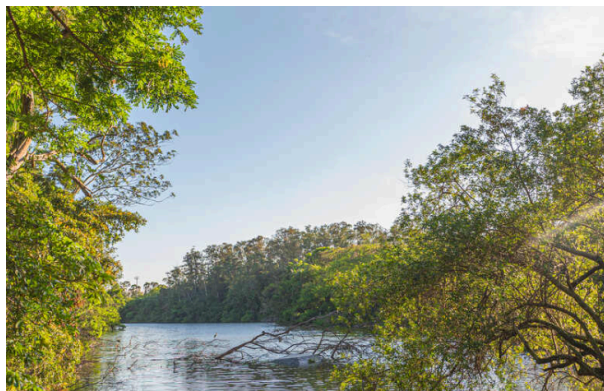
When consumers share resources, each reduces availability for others, creating indirect negative effects without direct contact.

Trophic Cascades

Predator effects on herbivores can benefit plants, creating indirect positive effects across food chains.

Facilitation-Competition Continuum

Relationships may shift from facilitation (nurse plants) to competition as species grow.



Nurse plants aid cactus establishment but may compete later, showing dynamic species interactions.

Mutualistic Interactions

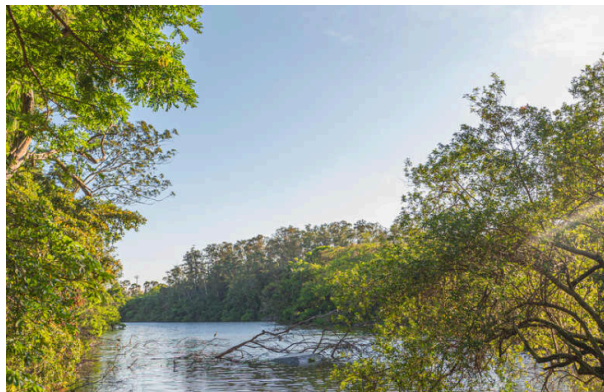
Species can collaborate for mutual benefit, with specialized adaptations.

Types of Mutualisms

- Trophic: Nutrient exchange (e.g., bacteria in ruminants)
- Defensive: Protection services (e.g., ants and acacias)
- Dispersive: Pollination and seed dispersal



Cleaning mutualisms between shrimps and fish provide parasite removal in exchange for food.



Ant-acacia mutualism involves obligate protection and nutrition exchange, with severe consequences for separation.

Evolutionary Implications

Mutualisms drive specialization and can become obligate, influencing community structure and ecosystem functioning.